

Mahdi Farahbakhsh

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Education

Texas A&M University

PH.D. IN COMPUTER ENGINEERING

- Advisors: Prof. Dileep Kalathil and Prof. Krishna Narayanan

College Station, TX, USA

2024 – Present (expected Aug. 2029)

Sharif University of Technology

M.S. IN COMPUTER ENGINEERING

- GPA: 19.31/20; ranked 1st in the Algorithms and Computations track

Tehran, Iran

2021 – 2023

Sharif University of Technology

B.S. IN COMPUTER ENGINEERING

- GPA: 17.90/20

Tehran, Iran

2015 – 2020

Research Interests

Diffusion models for images and language; reinforcement learning for LLMs; agentic AI; inference-time algorithms.

Publications

AgentDiscover: Autonomous Discovery with Minimal Search Scaffolding

[Under review](#) | [website](#) | [code](#)

MAHDI FARAHBAKHSH, ILAN SELA, FATEMEH DOUDI, VISHNU TEJA KUNDE, KRISHNA NARAYANAN, JEAN-FRANCOIS CHAMBERLAND, DILEEP KALATHIL

- Lets a coding agent plan its own search for discovery instead of following a hand-designed algorithm; matches or beats existing discovery frameworks on systems, mathematical-optimization and AtCoder heuristic-contest tasks at lower cost.

Inference-Time Search using Side Information for Diffusion-based Image Reconstruction

[NeurIPS 2026](#) | [arXiv](#) | [code](#)

MAHDI FARAHBAKHSH, VISHNU TEJA KUNDE, DILEEP KALATHIL, KRISHNA NARAYANAN, JEAN-FRANCOIS CHAMBERLAND

- Adds side information to pretrained diffusion inverse-problem solvers without retraining, via a plug-and-play inference-time search.

Reinforcement Learning for Diffusion LLMs with Entropy-Guided Step Selection and Stepwise Advantages

[NeurIPS 2026](#) | [arXiv](#) | [code](#)

VISHNU TEJA KUNDE, FATEMEH DOUDI, MAHDI FARAHBAKHSH, DILEEP KALATHIL, KRISHNA NARAYANAN, JEAN-FRANCOIS CHAMBERLAND

- Casts diffusion-LLM denoising as an MDP; policy updates focus on high-entropy steps, with stepwise advantages and no value network.

Experience

Texas A&M University

GRADUATE RESEARCH AND TEACHING ASSISTANT

- Research on diffusion models, RL for LLMs and discovery agents (see Publications); PyTorch, Hugging Face, multi-GPU SLURM training, W&B.
- Built the open-source AgentDiscover system: agent harness, Neo4j graph database and steering server (Python, Docker/Singularity).
- Teaching assistant for STAT 201 for two semesters: led weekly recitations and held office hours.

College Station, TX, USA

2023 – Present

Deep Finance

RESEARCH INTERN

- Led an intern team on an algorithmic global-macro FX strategy built from central-bank and statistical-agency data (Python).

London, UK (remote)

2023

Weadigit (stock-market data provider)

QUANTITATIVE RESEARCHER

- Built a portfolio-optimization pipeline from disclosed asset-manager holdings; beat the market in a two-year backtest (Python, SQL).

Tehran, Iran

2022 – 2023

Honors & Awards

2023 **Ranked 1st**, Algorithms and Computations track (M.S.), Sharif University of Technology

Tehran, Iran

2021 **Ranked 7th Nationally**, M.S. entrance exam in Computer Engineering (Konkur), ~20,000 candidates

Tehran, Iran

Professional Service

Reviewer

ICML 2026; Transactions on Machine Learning Research (TMLR)

Selected Coursework

Machine Learning & AI

Machine Learning; Deep Learning; Reinforcement Learning; Optimization for ML

Math & Statistics

Intro to Classical Analysis; Probability for Statistics; Bayesian Statistics

Algorithms & Theory

Data Structures and Algorithms; Design Algorithms; Randomized Algorithms